

Neural Networks and Deep Learning(NNDL)

Objective Question Bank

1. _____ proposed a learning law that explained how a network of neurons learned(Ans: Hebb)
2. _____ is a computer program that acquires, represents, and uses knowledge for a specific purpose .(Ans: Knowledge-based system)
3. A conventional computer program is characterized by _____ processing of data. (Ans: algorithmic)
4. _____ means the knowledge specific to the domain in which the problem is defined(Domain Knowledge)
5. _____ is the heuristic, experiential knowledge learned after solving a large number of problems(Surface Knowledge)
6. _____ refers to the basic laws of nature and the fundamental structural and behavioral principles of the domain.(Deep Knowledge).
7. What is object-level knowledge?
8. A _____ is a qualitative or quantitative description of an object or concept or event. (pattern).
9. Integration of symbolic AI and neural networks results in a so-called _____ (hybrid intelligent system)
10. A connection between nodes in different layers is called an _____ (interlayer connection)
11. The order of a neural network is the order of the _____ (highest order connection).
12. _____ refers to the case when a linear hyperplane exists to place the instances of one class on one side and those of the other class on the other side of the plane(Linear Separability)
13. In _____ all connections point in one direction i.e. from the input layer to output layer.(feedforward networks)
14. In recurrent networks, there are feedback connections or loops(True or False)(Ans: True)
15. A _____ perceptron consists of an input and an output layer(single-layer)
16. A multilayer perceptron is a feedforward neural network with at least one hidden layer(True or False)-(Ans: True)
17. The term _____ refers to the ability to learn and organize information without being given correct answers for input patterns.(self-organization)
18. What is the main problem of Hopfield network?(It settles into local minima by constant energy minimization)
19. What is Boltzmann Machine?
20. _____ can refer to either acquiring new knowledge or enhancing or refining skills.(Learning)
21. _____ learning assumes the availability of a teacher or supervisor who classifies the training examples into classes.(supervised)
22. _____ refers to learning based upon probability or statistical information of training instances. (statistical learning)
23. Rote learning is equivalent to learning by memorization(True or False)(Ans: True)
24. The _____ network is a multilayer feedforward network with a different transfer function in the artificial neuron.(Ans: backpropagation)

25. A _____ is a two-layer network whose output units form a linear combination of the basis(kernel) functions computed by the hidden units(Ans: radial basis function network)
26. What is statistical learning?
27. ID3 uses a _____(tree) representation for concepts.
28. What is version space
29. Reinforcement learning refers to a class of algorithms for learning automata(True or False)(Ans: True)
30. In _____ approach, the number of hidden units is determined by that of rules, and the number of hidden layers is determined by that of levels in the rule hierarchy.(Ans: rule-based connectionist network)
31. A system combining connectionists and symbolic reasoning models is an example of _____(hybrid intelligent system)
32. _____ uses a tree representation for concepts(Ans: ID3)
33. _____(Radial basis function) network is a two-layer network whose output units form a linear combination of the basis(kernel) functions computed by the hidden units.
34. Write the equation for activation level of hidden unit in radial basis function network?
35. Define rule-based neural networks
36. What is Knowledge-Based Conceptual Neural Networks(KBCNN)?
37. Write the equation for *Goodness of Fit*(G) in Constraint Based Neural Networks.
38. Deep Learning has become more useful as the amount of available _____ has increased(training data)
39. Deep learning has solved increasingly complicated applications with increasing _____ over time(accuracy)
40. Define rectified linear unit.[It is defined by the activation function $g(z)=\max\{0,z\}$]
41. _____ is used to compute the derivatives of functions formed by composing other functions whose derivatives are known(Ans: chain rule of calculus)
42. Write down the equation for logistic sigmoid activation function($g(z) = \sigma(z)$)(Refer Second Text book:Ian Goodfellow, Deep Learning page no 195)
43. What is the equation for hyperbolic tangent activation function(Ans: $g(z)=\tanh(z)$)